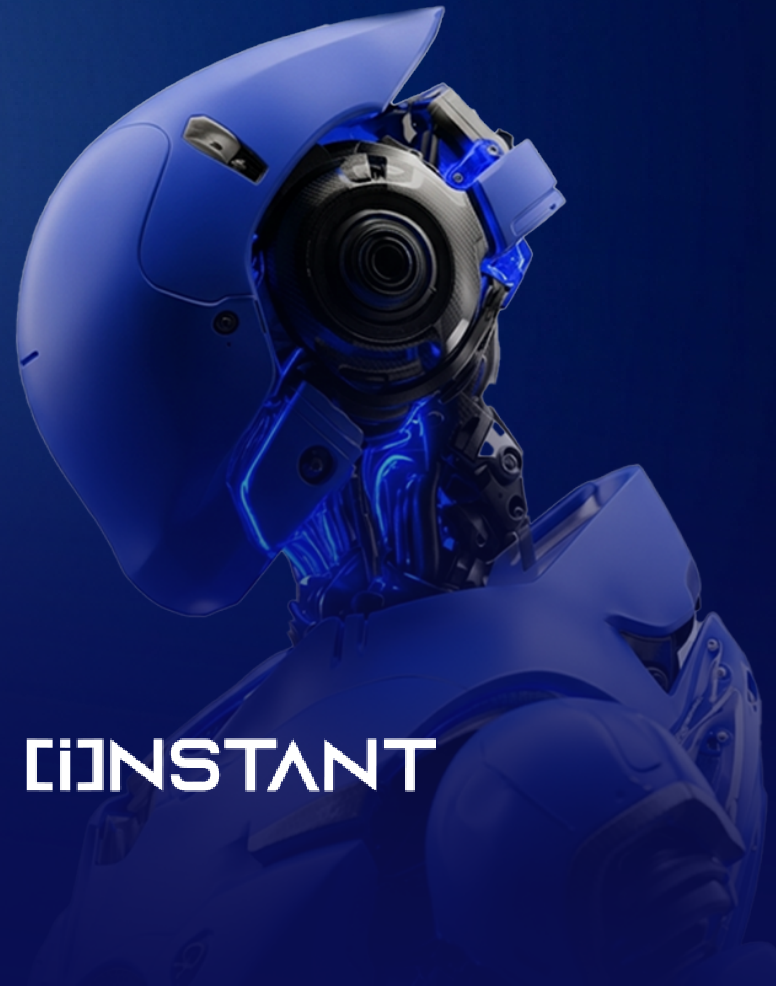
Orange
Digital Center

AI Hackathons

Hackathon executive presentation blueprint & comprehensive
Agenda



EXECUTIVE SUMMARY & CORE PHILOSOPHY

EVENT NAME	AI Hackathons
ORGANIZED BY	CREATIVA Innovation Hubs, ITIDA, TIEC, Orange Digital Center, INSTANT Software Solutions
DATE	August 16, 2026 – August 20, 2026 (5 Days)
CORE THEME	Building a safe, evidence-grounded, traceable, and trustworthy Retrieval-Augmented Generation (RAG) system using official medical guidelines.

Core Philosophy: **Fluent Answer ≠ Safe Answer.** Clinical decision support must be grounded in official evidence with explicit citations, transparent retrieval, and verified refusal logic.

EVENT SCHEDULE & DELIVERY FORMAT

DAY	DATE	FORMAT	CORE FOCUS	KEY OUTPUT
Day 1	Sun, Aug 16	Offline	Research, Scope & Ingestion, Team Formation	Searchable Vector DB with Metadata
Day 2	Mon, Aug 17	Offline	Retrieval Optimization	Measured Baseline & Precision@K
Day 3	Tue, Aug 18	Online	Grounded Generation & Citation	Structured, Cited RAG Pipeline
Day 4	Wed, Aug 19	Online	Safety, Guardrails & Internal Evaluation	Guardrail Workflow & Benchmark Dashboard
Day 5	Thu, Aug 20	Offline	Final Presentation & Judge Evaluation	Frozen Prototype, Live Demo & Pitches

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CLINICAL PROBLEM & SCOPE

THE RISK

Generic LLMs answer from parametric memory — fluent, confident, and often ungrounded. In a clinical context, that risk of hallucination is unacceptable.

Fluent ≠ Safe. Every clinical recommendation must trace back to an official, citable source.

OUR SCOPE

A narrow, well-defined clinical topic — e.g. Adult Hypertension Management, Diabetes Screening, or Asthma Guidance — sourced strictly from official, public guidelines (WHO, CDC, NICE, USPSTF).

No private or credential-gated data. Every answer is grounded, cited, and traceable to a page and section.

END-TO-END SYSTEM ARCHITECTURE

01 **Ingestion**

02 **Chunking**

03 **Embeddings**

04 **Retrieval**

05 **Guardrails**

06 **Grounded LLM**

07 **Evidence Panel**

A modular, layered pipeline — each stage independently testable, each output auditable.

INGESTION & CHUNKING STRATEGY

1 Data Sourcing

Official, public guideline PDFs only — WHO, CDC, NICE, USPSTF. No private or credential-gated data.

2 PDF Cleaning

Extract text while preserving page numbers; strip headers, footers, and extraction artifacts.

3 Section-Aware Chunking

400–800 token chunks that preserve recommendation context and section boundaries.

4 Metadata Schema

document_name, page_number, section_title, chunk_id, and source_url stored with every vector entry.

RETRIEVAL PIPELINE & OPTIMIZATION

KEY EXPERIMENTS

- Top-K Tuning — Top-3 (precision), Top-5 (balanced), Top-10 (recall)
- Chunk size & overlap — 400–600 vs. 700–900 tokens
- Semantic Search vs. Keyword (BM25) vs. Hybrid vs. Reranking
- Mini evaluation set of 15–20 labeled clinical test questions
- Evidence Panel UI — shows retrieved chunks, scores & metadata before generation

PRECISION@K

$$\frac{\text{Relevant Chunks in Top-K}}{K}$$

Measured on the day-2 evaluation set and reported for Precision@3 and Precision@5, alongside documented failure-mode fixes.

GROUNDED GENERATION & CITATION MECHANICS

The retrieved guideline text is the absolute source of truth — the LLM acts strictly as an evidence synthesizer, never a diagnostician.

Recommendation

Direct, concise summary

Supporting Evidence

Bullet points with quoted excerpts

Citations

Document + Section + Page Number + Chunk ID

Confidence & Safety

High / Medium / Low / Insufficient Evidence + Disclaimer

SAFETY & GUARDRAIL WORKFLOW

01

Input Risk Classification

Queries are categorized as Allowed, Needs Caution (patient scenarios), or Refuse/Redirect (emergencies, out-of-scope).

02

Retrieval Confidence Thresholds

Generation is blocked or confidence downgraded when similarity scores fall below defined cutoffs.

03

Unsupported Claim Detection

Generated outputs are post-processed to verify every claim against the retrieved chunk text.

EMPIRICAL EVALUATION DASHBOARD

Benchmarked across 20–30 structured test cases: direct, multi-chunk, ambiguous & out-of-scope.

Retrieval Precision@K

Relevant Chunks in Top-K ÷ K

Citation Accuracy

Correct Supporting Citations ÷ Total Citations Checked

Faithfulness / Unsupported Claim Rate

Unsupported Claims ÷ Total Claims Generated

JUDGING CRITERIA & RUBRIC BREAKDOWN

CATEGORY	WEIGHT	KEY EVALUATION POINTS
Retrieval Quality	**30%**	Relevant chunk extraction, Precision@K scores, chunk metadata transparency, vector index structure.
Grounding & Citation	**25%**	Accurate page/section citations, zero unsupported claims, minimal hallucination, tight claim-to-chunk binding.
System Architecture	**15%**	Modular pipeline design, clear separation of layers, clean architectural diagram.
Evaluation Depth	**15%**	At least 2 quantitative metrics reported (20+ Q dataset), clear failure analysis and trade-off understanding.
Safety & UX	**15%**	Input guardrails, safe refusal logic, confidence indicators, clinical disclaimer, smooth live presentation & demo.

LIVE PROJECT DEMO

Three predefined query scenarios, executed live in front of the judges.

CASE A

SUCCESS

Single / multi-chunk query answered with exact page-level citations.

CASE B

COMPLEX MULTI-STEP

Multi-section synthesis, returned with fully structured supporting evidence.

CASE C

SAFE REFUSAL

Out-of-scope query correctly triggers an insufficient-evidence refusal.

FINAL RESULTS, ROADMAP & Q&A

1

Key Achievements

Presented live on Day 5 — frozen prototype, benchmark scores, and judge scoring.

2

Clinical Safety Disclaimer

This system supports — never replaces — clinical judgment. Outputs are guideline-grounded, not diagnostic.

3

Scalability Roadmap

Path from a single clinical topic to a multi-guideline, continuously-validated knowledge base.

THANK YOU.

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